

TEST BANK

CHAPTER 6

INTERTIDAL ECOLOGY

Matching

In this section, you are to select from the choices listed in column B the term or phrase that best matches the term or phrase listed in column A and then to place the letter for the selected choice from column B in the space provided to the left of the corresponding item in column A.

[Note to instructors: Here, the choices are correctly matched and thus should be randomly assorted before the exam is prepared.]

Column A		Column B
___ 1. diffuse predation	—————>	A. no single predator dominates the community
___ 2. weak predation	—————>	B. the total effect of predation is small
___ 3. diurnal tide	—————>	C. single low and high tide per day
___ 4. semidiurnal tide	—————>	D. two high and two low tides per day
___ 5. critical tide level	—————>	E. increased exposure during low tide
___ 6. keystone species	—————>	F. organisms with the greatest influence on community structure
___ 7. <i>K</i> -selected	—————>	G. slow growing and long lived
___ 8. <i>r</i> -selected	—————>	H. fast growing and short lived
___ 9. swash	—————>	I. water that runs up the beach after the wave breaks
___ 10. seep	—————>	J. a region where water drips down from above
___ 11. frondose	—————>	K. leafy growth form
___ 12. planktotrophic larvae	—————>	L. feed while swimming
___ 13. lecithotrophic larvae	—————>	M. do not feed while swimming
___ 14. bet hedging	—————>	N. algae with two different growth forms
___ 15. insolation	—————>	O. amount of sunlight reaching Earth
___ 16. patch	—————>	P. high localized distribution of organisms
___ 17. Stephenson Universal System	—————>	Q. zonation patterns on rocky shores
___ 18. reflective beach	—————>	R. wave action hits the beach face directly
___ 19. dissipative beach	—————>	S. wave action is spread out over a flat region
___ 20. backwash	—————>	T. water flowing down a beach slope

Multiple Choice

Circle the choice that best completes or answers the given questions.

[N.B.: The correct answer to each of the questions is indicated by an asterisk. For most of the questions, the letter "a" is the correct choice; reorder the choices as you see fit.]

- Which one of the following is a major sea that does not experience tidal action?
 - Baltic Sea *
 - Arctic Sea
 - Gulf of Mexico
 - Bay of Fundy
 - Sea of Cortez
- Which one of the following terms is used to describe a tidal cycle with a single high and low per day?
 - Diurnal tides *
 - Semidiurnal tides
 - Mixed tides
- Which one of the following terms is used to describe a tidal cycle having two highs and two lows of about the same size per day?
 - Diurnal tides
 - Semidiurnal tides *
 - Mixed tides
- Which one of the following terms is used to describe a tidal cycle having two high and two low tides of unequal height per day?
 - Diurnal tides
 - Semidiurnal tides
 - Mixed tides *
- Barnacles attach to hard substrata by which of the following substances?
 - Cement *
 - Byssal threads
 - Long spines
 - A long plantar surface (foot)
 - None of the above
- Which of the following organisms avoid the mechanical stress of wave action by permanently attaching themselves to hard substrata?
 - Barnacles, such as *Balanus*
 - Oysters, such as *Crassostrea*
 - Mussels, such as *Mytilus*
 - Crabs, such as *Callinectes*
 - More than one of the above *
- Which of the following is a strategy frequently absent from intertidal animals?
 - Osmoregulation *
 - Vascularization of the mantle cavity
 - Proliferation of blood vessels of the integument
 - Permanent attachment
 - Strong enlarged ventral foot
- The ____ describe(s) zonation along rocky shores as being composed of three main divisions, known as the supralittoral, midlittoral, and infralittoral regions.
 - Stephenson Universal System *
 - critical tide level
 - alternative stable states
 - None of the above
- Points along the vertical profile of a shore exhibiting pronounced increases in exposure time (time spent out of the water) over short distances are called _____.
 - critical tide levels *
 - mixed tides
 - diurnal tides
 - semidiurnal tides
 - More than one of the above
- Which of the following does not appear to be an important biotic controlling factor of zonation in the rocky intertidal zone?
 - Parasitism *
 - Competition
 - Predation
 - Grazing
 - Larval settlement
- In the rocky intertidal zone, which one of the following resources is in limited supply?
 - Space *
 - Food
 - Mates
 - Vitamins
 - Oxygen

12. Which one of the following is the keystone species of the rocky intertidal zone of the Pacific Northwest?
- The starfish *Piaster* *
 - The mussel *Mytilus*
 - The gastropod *Nucella*
 - The barnacle *Balanus*
 - The horseshoe crab, *Limulus*
13. Which one or more of the following features are an adaptation by algae that reduces grazing pressure?
- Deposition of calcium carbonate in their tissues
 - Deposition of noxious compounds in their tissues
 - Modification of growth form
 - Growth to a large size
- 1 only
 - 2 only
 - 1 and 2
 - 2, 3, and 4
 - All of the above *
14. Which of the following organisms occurs in large clumps of genetically identical individuals (clones)?
- The sea anemone *Anthopleura* *
 - The mussel *Mytilus*
 - The barnacle *Chthamalus*
 - The gastropod *Littorina*
 - None of the above
15. The creation of open bare patches in any area of the rocky intertidal zone results in rapid colonization by ____ species.
- opportunistic [fugitive] *
 - K*-selected
 - keystone
 - deposit feeder
 - None of the above
16. Which one or more of the following features are major contributors to zonation in sandy beaches?
- Space
 - Competition
 - Predation
 - Wave action
 - Particle size
 - Beach slope
- 1 and 2
 - 1, 2, and 3
 - 4 and 5
 - 4, 5, and 6 *
 - 1 and 6
17. Which of the following occurs in the redox-potential discontinuity layer?
- Eh changes rapidly from positive to negative. *
 - Eh changes rapidly from negative to positive.
 - Reduced compounds rapidly decrease in concentration.
 - The density of aerobic bacteria increases.
 - The density of aerobic animals increases.
18. Which one of the following compounds is most oxidized?
- Fe^{+3} *
 - Fe^{+2}
 - H_2S
 - CH_4
 - NH_4^+
19. In anaerobic respiration, the final hydrogen–electron (H^+/e^-) acceptor may be any of the following molecules except which one?
- Oxygen *
 - Carbon dioxide
 - Nitrate
 - Nitrite
 - Sulfur
20. The Eh value of an anoxic mud would most likely ____ mV.
- be between 0 and -100 *
 - be $< +300$, but $> +200$
 - be $< +400$, but $> +300$
 - approach $+400$
21. As one proceeds from an oxidized zone through the RPD layer and into the anoxic zone, the color of the mud changes from ____.
- brown to gray to black *
 - black to gray to brown
 - gray to brown to black
 - brown to black to gray
 - gray to black to brown
22. The major predators on mud flats are ____ and ____.
- fish, birds *
 - fish, crabs
 - fish, snails
 - crabs, snails
 - crabs, birds

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23. A central theme in marine ecology is that whenever predation is reduced, ____.
- competition is increased *
 - competition is decreased
 - grazing pressure is decreased
 - densities are reduced
24. An osmoconforming organism is ____.
- an organism in which the mechanism to control the salt content is poorly developed *
 - an organism in which the mechanism to control the salt content is highly developed
 - an animal that conforms to the contours of the substratum
 - an animal that does whatever the other animals surrounding it does
 - more than one of the above
25. The Earth and moon form an orbiting system, revolving around ____.
- a common center that is not the center of mass of the Earth *
 - the center of mass of the Earth
 - the center of mass of the moon
 - the center of mass of Mars
 - the center of mass of Jupiter
26. The position of a satellite when it is at its closest point to the planet it orbits is called the ____.
- perigee *
 - apogee
 - geewiz
 - Cheese Whiz®
 - None of the above
27. A group of genetically identical sea anemones composes a ____.
- clone *
 - sere
 - swash
 - berm
28. A rapid decrease in oxygen concentration within a muddy sediment leads to a ____.
- rapid drop in the redox potential *
 - rapid rise in the redox potential
 - conversion of bacterial metabolism over to aerobic respiration
 - change in the color of the mud to a much lighter hue
29. Highly specialized tentacles in sea anemones are called ____.
- acrorhagi *
 - clones
 - bars
 - berms
 - seres
30. There is little primary productivity on open sand beaches; the animals found there belong to each of the following feeding types except which one?
- Predators *
 - Filter feeders
 - Detritus feeders
 - Scavengers
31. Each of the following except ____ is a dominant animal on sand beaches and sand flats.
- large crabs *
 - polychaete worms
 - crustaceans
 - bivalve mollusks
32. Cobble beaches occur in North America in all of the following places except ____.
- the Chesapeake Bay *
 - the Long Island Sound
 - Narragansett Bay
 - Puget Sound
 - Maine estuaries

True/False

Mark each of the given statements with the letter *T* if it is true and *F* if it is false.

[Instructors: In some cases, a few options are included for the same question.]

- [False]* The Earth and moon form an orbiting system, revolving around a common center that is the center of mass of the Earth.
- [True]* Except for the scale, zonation seen in rocky intertidal regions is similar to the zonation pattern seen in mountain habitats.
- [True]* A major abiotic difference between slopes facing south and those facing north is the amount of insolation received by each.

4. [False] Younger, less dense clumps of mussels appear to be more vulnerable to wave action than older, more dense clumps.
5. [True] The sea anemone, *Anthopleura elegantissima*, exhibits territoriality when adjacent, but distinct, clones fight when placed next to one another.
6. [True] Fine sand can hold more water in the interstitial spaces than coarse sand.
7. [False] Coarse sand can hold more water in the interstitial spaces than fine sand.
8. [True] Backwash is the term used to describe the water that flows back down the beach after a wave breaks.
9. [True] Swash is the water that runs up a beach after a wave breaks; this action carries particles with it which may remain on the beach.
10. [True] In a dissipative beach, the sediments are fine, wave action is strong, and the beach slope is gentle.
11. [False] In a reflective beach, the sediments are coarse, wave action is minimal, and the beach slope is steep.
12. [True] Large plants are absent from sandy beaches and sandy flats.
13. [True] There is very little primary production on open sand beaches.
14. [False] Sand beaches have some of the highest levels of primary production in the inshore region.
15. [True] Eh is inversely proportional to the concentration of reduced inorganic compounds.
16. [False] Eh is directly proportional to the concentration of reduced inorganic compounds.
17. [False] As the concentration of reduced inorganic compounds decreases, Eh becomes more negative.
18. [True] The reason anoxic muds are black is that hydrogen sulfide generated by decomposition reacts with available iron to form iron sulfide, a black precipitate.
19. [True] The chemosynthetic bacteria in the RPD that use hydrogen sulfide as a source of energy possess the same metabolic pathways of the primary-producer bacteria present in the submarine thermal hot vents (hydrothermal vents).
20. [True] Mud flats are unusual in that they possess two different layers where primary production may occur: the surface, where photosynthesis occurs, and a deeper layer, where chemosynthesis occurs.
21. [True] Under pressure, a saturated sandy sediment (gel) will become a liquid.
22. [False] Coarse-grained sediments can hold more water than fine-grained sediments.
23. [False] Ice is an insignificant feature on the sand-flat communities of the higher latitudes.
24. [False] Few intertidal animals have planktonic larvae.
25. [True] The breeding cycles of most intertidal animals are synchronized with the tides.
26. [False] Throughout the intertidal regions of the world's oceans, we find that the water temperature always has a greater range than the local air temperature.
27. [True] Spring tides occur when the sun and the moon are directly aligned.
28. [True] The intertidal zone encompasses the smallest of all marine systems.
29. [False] The intertidal zone has many terrestrial plants and animals.
30. [False] Considerable primary production occurs in the open sandy intertidal zone.
31. [True] Some algal species have more than one growth form.
32. [False] Many mud-flat infaunal animals are unable to tolerate low-oxygen conditions.
33. [True] Sand dollars accumulate iron compounds in a special area of their digestive tracts; this extra weight helps to keep them in place during times of wave action.
34. [False] A central theme of intertidal ecology is that whenever predation is reduced, competition will be decreased.
35. [True] Because of the ceaseless movement of the surface layers of sediment on sand beaches, few large organisms occupy the surface.

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Short Answer (Fill in the Blank)

1. The point in the orbit of the moon when it is closest to the Earth is called the perigee.
2. The point in the orbit of the moon when it is farthest from the Earth is called the apogee.
3. The redox-potential discontinuity layer represents a region where anaerobic and aerobic layers meet.
4. Extreme high tides take place in basins with unusual morphometry (geometry), such as the Bay of Fundy or Cook Inlet.
5. Mussels such as *Mytilus* use byssal threads to keep themselves attached to hard substrata.
6. The Stephenson Universal Scheme, which describes zonation along rocky shores, is composed of three main divisions.
7. The three main divisions of the Stephenson Universal Scheme of zonation are the supra-, mid-, and infra- littoral zones.
8. Points along the vertical profile of a shore exhibiting pronounced increases in exposure time (time spent out of the water) over short distances are called critical tide levels.
9. In the rocky intertidal zone, the most important limiting resource is space.
10. Most algae grow best in the lower intertidal regions, but are limited because of grazing.
11. The presence of a home scar or depression at a specific site in the home range of the limpet, *Collisella digitalis*, permits this intertidal mollusk to survive when the tide recedes.
12. The importance of particle size to organisms in a sandy beach lies within the effect of particle size on water retention.
13. Through the phenomenon of capillary action, fine sand can hold more water than coarse sand.
14. Defining wave action and particle size fixes the third variable, slope.
15. Defining slope and particle size fixes the third variable, wave action.
16. Swash is the term used to describe the water that runs up the beach after a wave breaks.
17. A dissipative beach occurs where wave action is strong and the sediments are fine. The slope of such a beach is flat and eroded.
18. A reflective beach is more fully developed where wave action is minimal and the sediment is coarse. In such a beach, there is no surf zone, and the beach slope is steep.
19. Between the upper aerobic sediment layer and a lower, anaerobic zone is a transition zone called the redox-potential discontinuity (RPD) layer.
20. The physical phenomenon of thixotropy allows a mating pair of *Limulus polyphemus* to burrow into wet sand fairly easily. [Instructors will need to provide additional information in lecture.]
21. Barnacles are arthropods that are important members of the fouling community of docks, boats, and other hard substrata.
22. The life zone on a sandy beach that begins at the high-tide line and extends to the low-water mark of low tide is called the intertidal zone.
23. An Eh value of more than 200 mV indicates that some oxygen is present.
24. Acrorhagi are specialized tentacles on certain kinds of sea anemones that are used as a type of weapon by members of opposing clones.
25. Skates, rays, and flatfishes are significant predators of infaunal animals on sand and mud flats.
26. There are no sharp boundaries between what constitutes a sandy shore and a muddy shore; they are simply at opposite ends of a continuum.
27. Tide pools may be considered to be refuges for mobile intertidal animals of the rocky shore.
28. Algae have several defenses against grazing animals; one involves laying down the mineral calcium carbonate in their tissues.

29. Beaches are defined by particle size, wave action, and slope.
30. Predation on intertidal organisms by birds may be very intense at certain times of the year and much less intense at other times because of their migratory patterns.
31. The bacteria in the RPD layer include chemolithoautotrophic, or sulfur, bacteria, which use sulfur compounds as their source of energy.
32. A keystone species is a single predator that controls the abundance of a prey species that, in turn, could exclude all other species and dominate the community.
33. The involvement of more than a single predator in preventing competitive domination is called a diffuse predation. If predation alone cannot control the abundance of the competitive dominant, then the situation is called weak predation.
34. The types of organisms whose absence is conspicuous on open sand beaches are large plants and sessile animals.

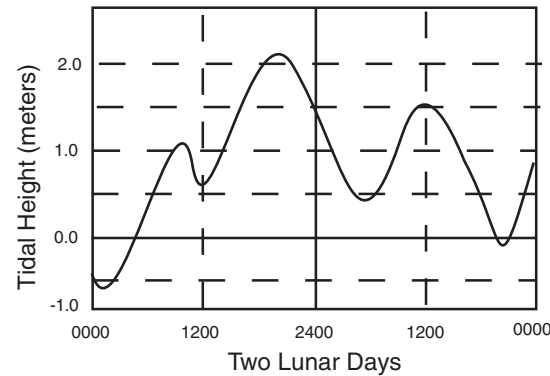
Essay Discussion Questions

Develop complete answers to these questions.

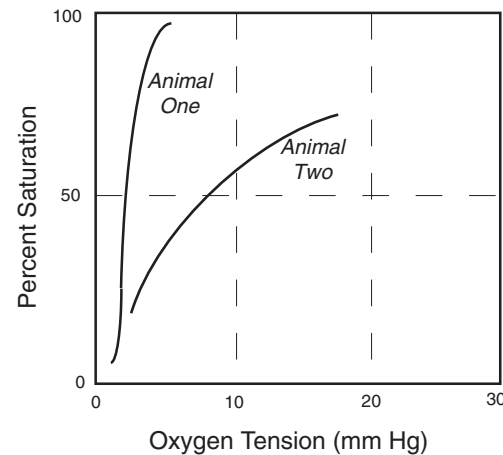
1. Why is the intertidal zone the most well-known marine habitat? How is it important to the environmental movement? Discuss.
2. Compare and contrast the critical tide level in two intertidal regions, one with diurnal tides and the other with mixed tides.
3. How do intertidal animals minimize exposure to the stresses of their habitats? Discuss.
4. How could water seeps alter a vertical zonation pattern in the rocky intertidal zone? Discuss.
5. How do intertidal animals that are exposed to the sun's radiation avoid overheating? Discuss several different adaptations.
6. How does the physics concept of blackbody radiation relate to the heat balance of intertidal animals?
7. Discuss what led the Stephensons' to propose their universal scheme and why it is useful for researchers.
8. Discuss why researchers have had difficulty finding a good correlation between the critical tide level and distribution of organisms.
9. Why is there little overlap in the distribution of adult barnacles of the genera *Chthamalus* and *Balanus* on the rocky intertidal shores of Scotland? Discuss the biotic and abiotic factors controlling their distribution.
10. How can natural disasters, and even human blunders, be useful to marine biologists? Discuss, citing examples to support your essay.
11. Discuss the biogeographical importance of grazing from the tropics towards the poles.
12. How do cracks and crevices influence the rocky intertidal community? Discuss.
13. What abiotic factors are subject to great variation in tide pools, and how do they affect the organisms of that habitat? Discuss.
14. Why must a researcher who is studying zonation on a rocky shore take samples from the same spot at different times of the year over the course of the study? Discuss.
15. How can storms influence the distribution of organisms in the sandy intertidal zone? Discuss.
16. If space is the dominant limiting factor for organisms living in the rocky intertidal zone, what is (are) the dominant limiting factor(s) for organisms living in sandy beaches?
17. What do the oxygen dissociation curves for oxygen-carrying molecules of animals living in the mud flats indicate?
18. Draw a well-labeled figure on an x - y axis showing the oxygen dissociation curves for two hypothetical animals: one that lives on top of the mud and one that burrows 10 cm into the mud.

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19. Using the following hypothetical tidal curve, approximate the maximum time of continuous submergence for the depths of 0.5 and 1.5 m:



20. Using the following oxygen dissociation curves, determine which of the two hypothetical animals is able to extract oxygen from its environment at lower oxygen concentrations, making sure to defend your answer:



21. The abundance of birds foraging on intertidal flats may be quite high, suggesting that the birds will deplete the food resources or are subject to intense competition. Discuss the reasons that tidal-flat birds neither severely deplete their resources nor are subject to severe competition.
22. What is a critical tide level (CTL)? Discuss three organisms to which the CTL is of obvious importance.
23. Name and discuss two different mechanisms that may be observed among intertidal organisms that allow them to maintain their heat balance (that is, not become so hot that they overheat and die).
24. Bacteria are very important in the intertidal community. Name two different roles that bacteria play in the sand and mud of the intertidal shore.
25. What accounts for the variation in sediment particle size in various parts of the shallow-water zone? Why might one come across regions where shells of dead invertebrates (for example, clams, oysters, and snails) can be found in abundance? Why are these areas sometimes dominated by one shell type, while in another area, a different species predominates? (*Hint*: Why do leaves tend to settle in certain spots around buildings in the fall? Why does dust tend to collect in greater amounts in certain parts of your room?)
26. Discuss the following statement: The intertidal zone is an extension of the marine environment and is inhabited almost exclusively by marine organisms.
27. Why are there so few terrestrial organisms that exploit the marine intertidal habitat?
28. What is a critical tide level? Suggest several marine organisms that have an obvious critical tide level. Support your reasoning for offering those organisms.

29. How can air temperature and the tidal cycle team up to drastically influence organisms living in the intertidal zone?
30. How do the intertidal organisms avoid death by desiccation? Discuss several mechanisms.
31. Suggest two different functions for a thick shell in an intertidal gastropod, and compare those functions with those of an intertidal bivalve that is buried in the mud flat.
32. What is the ecological significance of osmoconformers?
33. What is the genetic consequence of a sedentary organism such as a gastropod or bivalve that has a larval stage which has only a limited planktonic existence? Would you expect this organism to have a wide geographic distribution? Why or why not? Which of the two types of larva—those with a limited or those with an extensive planktonic existence—would be able to become more widely distributed given favorable conditions?
34. The rocky shores are the most densely populated of all the intertidal regions. They also have the greatest diversity of animal and plant species and stand in striking contrast to the almost barren appearance of sand and mud shores. What ecological factor(s) would account for this dense population and high diversity? Discuss.
35. Is the Stephenson Universal Scheme realistic for all intertidal zones? Discuss.
36. Is zonation found only on rocky shores? What sorts of organisms are found in zones along shorelines that are mud or sand flats? What is (are) the principal ecological factor(s) controlling zonation in the intertidal zone?
37. What physical factors are important in producing zonation of intertidal organisms?
38. What abiotic factors are important in producing zonation of intertidal organisms?
39. How might the compass direction of the slope of a rocky shore be very important in structuring the zone?
40. Discuss how UV light limits organisms on the intertidal shore.
41. Discuss the biological factors that are important in producing zonation of intertidal organisms.
42. Discuss the importance of space and surface area in the marine rocky intertidal zone. Use specific examples from the literature.
43. How are the communities of a tide pool and a bromeliad similar? How are they different? Discuss.
44. What roles do refuges play in structuring the rocky intertidal zone?
45. What behaviors does a chiton have that make it a superior competitor for space?
46. In which phyla does territoriality play an important role in structuring the rocky intertidal zone? What sorts of behaviors or life-history strategies are exhibited by these organisms?
47. How do larvae help create the community structure of the rocky intertidal zone?
48. Draw a hypothetical age–frequency histogram, with frequency on the vertical (dependent) axis and age classes on the horizontal (independent) axis.
49. How do tide pools vary in their salinity, temperature, and oxygen concentration as a function of time and tidal cycle?
50. How are wave action, particle size, and water retention related? How does particle sorting take place?
51. What adaptive features help keep the sandy-beach infauna in the sand?
52. What problems do sand and mud offer the infauna?
53. Are carnivores rare in mud-flat habitats? Discuss.
54. What sorts of zonation patterns do intertidal beaches display?
55. What is the dominant feeding type(s) of the infauna in mud flats? To what phyla do these organisms belong?
56. Which term, “community” or “continuum,” is best for describing the transition from a mud flat to a sand flat? Discuss.

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57. How do intertidal organisms reduce heat gain from the environment? Discuss.
58. Discuss how intertidal organisms reduce mechanical stress caused by wave action.
59. What problems would an intertidal animal encounter were it to feed only during a rising or falling tide?
60. What important differences are there between rocky shores of the Atlantic and the Pacific?
61. What physical and chemical factors affect the organisms of a mud flat (as contrasted with a sandy beach)?
62. Describe the oxygen concentration and Eh from the water–surface interface to a depth of 20 cm into the sediment in a mud flat. Include in your essay a figure of a vertical profile of the oxygen concentration and Eh values. Discuss why it looks this way.
63. Discuss how one could, without using sophisticated chemical probes, determine the approximate level at which the oxygen concentration in a sediment fell to zero. What are the biological consequences of the transition from oxic to anoxic conditions? Discuss.
64. How will the oxygen concentration in a mud flat influence the vertical distribution of the following organisms in the mud: bacteria, algae, protozoans, various worms, clams, and snails?
65. Why do macroalgae grow well on mud flats, but not on sand beaches?
66. What adaptations allow organisms to survive on an exposed rocky shore? Discuss, giving examples to support your ideas.
67. Is space limiting in the rocky coast?
68. What have been the problems with the use of the concept of the keystone species? Discuss.
69. Support or refute the following statement: Whenever predation is reduced, competition will be increased.
70. Why is larval recruitment variable over space and time?
71. Why is it that the age structures of the dominant organisms in the intertidal zone are mainly of a single-year class? (N.B.: Depending on the species, this single-year class may vary.)
72. If predation of shorebirds on intertidal organisms is very intense, why aren't these communities overexploited by predation? Discuss.
73. Discuss how the cage-experiment technique demonstrates that *Littorina littorea* controls algal growth on marine cobble beaches.
74. What are the important factors that define beaches, and how are they interrelated?
75. Discuss the concept that sand and mud shores represent opposite ends of a continuum. What does this conception mean to the various organisms that may be found to occupy their own part of this continuum?
76. Discuss how temperature varies within the intertidal habitats of mud, sea-grass stands, and a rocky zone.
77. Discuss how the birds of the tidal flats in New England divide up the habitat.
78. How does the tidal cycle interact with the diel cycle of light and dark to influence the behavior of the semiterrestrial isopod *Tylos*?
79. Distinguish between diffuse and weak predation.
80. What is bet hedging, and how does it affect the survival of a species?
81. In what ways is territoriality manifested in the intertidal zone?
82. Why are cobble beaches fundamentally different from other coastlines? Provide examples of the differences.
83. Where are cobble beaches found in North America, and why are they distributed in this way?
84. How do the actions of waves provide both positive and negative affects for the organisms of the intertidal zone?
85. Speculate as to what a rocky coast might look like if there were no moon.
86. Challenge Question: Discuss the genetics of clonal intertidal invertebrates. How is a knowledge of self important?

87. Challenge Question: Scallops are dark on one side and light on the other. What might be the adaptive significance of this feature? Discuss possible working hypotheses.
88. Challenge Question: How might the critical tide level be similar to the distribution of moss around the circumference of a tree? Discuss.
89. Challenge Question: Does the structure of the rocky intertidal zone support the ecological theory of community or continuum? Discuss.
90. Challenge Question: Compare and contrast the age structure of organisms in a rocky intertidal zone and in a mud flat.
91. Challenge Question: Compare the horizontal banding pattern (zonation) of organisms in the marine intertidal zone with the zonation in a temperate forest (*a tropical forest*).
92. Challenge Question: Define the concept of the keystone species, providing examples. Invent a metaphor for this concept from our modern world.
93. Challenge Question: What is the importance of the redox-potential discontinuity (RPD) layer to the decomposition of organic material?
94. Challenge Question: What is the importance of the redox-potential discontinuity (RPD) layer to the organisms that inhabit the marine intertidal zone?
95. Challenge Question: Why haven't starfish and sea urchins invaded freshwater areas?
96. Challenge Question: Considering the theories of plate tectonics and evolution, explain why the number of species and species diversity are greater in the Pacific intertidal zone than in the Atlantic intertidal zone.

